10.6 Segment Relationships in Circles

Name _____ Period ____ Date _____

Geometry 10.6 (30-minute plan) · Student sheet

I can…

- find lengths of segments of chords that cross inside a circle.
- find lengths of secant segments that meet outside a circle.
- find lengths when a tangent and a secant meet outside a circle.

Vocabulary · fill in the missing word

Segments of a chord — the two _____ of a chord that is cut by another chord

Tangent segment — a segment that is tangent to a circle at one of its _____

Secant segment — a segment that _____ a chord and has exactly one endpoint outside the circle

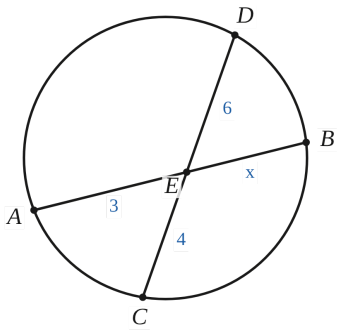
External segment — the part of a secant segment that is _____ the circle

Part 1 · Notes and examples (fill in each note, then work the example)

Segments of Chords Theorem: if two chords cross inside a circle, the _____ of the lengths of the segments of one chord equals the product for the other chord: EA · EB = EC · ED.

Example 1

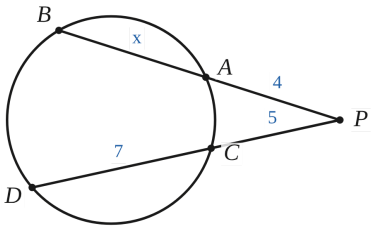
Chords $\overline{AB}$ and $\overline{CD}$ cross at E. Find the value of x.



Segments of Secants Theorem: if two secants share an endpoint outside a circle, then (external segment) · (_____) is the same for both.

Example 2

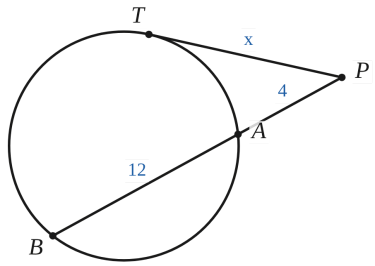
Find the value of x.



Segments of Secants and Tangents Theorem: if a tangent and a secant share an endpoint outside a circle, then the _____ of the tangent segment = (external segment) · (whole secant segment).

Example 3

$\overline{PT}$ is tangent to the circle at T. Find the value of x.



Part 2 · Partner practice

Partner practice

A tangent segment is 12. A secant from the same point has an external segment of 8 and an internal segment of x. Find x.